Accelerated Math II/III Name________________________________

HW # 7 Review #2 Part 1

Date________________ Period____

Sketch the graph of each line.

1) $8x + 5y = -25$

2) $2x - y = 0$

Graph each function using a minimum of 5 points on graph paper. Make a table of values to show your work. Remember, parabolas are symmetric. Identify the vertex, axis of symmetry, intercepts, interval of increase and interval of decrease, domain and range.

3) $y = -x^2 - 4x$

4) $y = x^2 - 4x + 3$

5) $y = (x + 3)^2 - 4$

6) $y = 2(x - 3)^2 + 1$

Graph each function.

7) $g(x) = \begin{cases} -x + 1, & x \le -3 \\ -|x|, & -3 < x \le 2 \\ \dfrac{|x|}{2}, & x > 2 \end{cases}$

8) $g(x) = \begin{cases} -4, & x \le -4 \\ -2x - 2, & -4 < x \le 2 \\ x + 2, & x > 2 \end{cases}$

Graph each equation.

9) $y = 3|x + 3| - 2$

10) $y = -2|x - 1| - 2$

Solve each equation.

11) $10 + 8|5x - 2| = 34$

12) $7|5 - 2n| - 1 = 6$

Solve each inequality. Write your answer in interval notation.

13) $10|-9 + 2x| - 9 < 61$

14) $9|9p - 6| + 6 \le 114$

Write the slope-intercept form of the equation of each line given the slope and y-intercept.

15) Slope $= -5$, y-intercept $= -5$

Write the slope-intercept form of the equation of each line.

16) $5x + 4y = 8$

Write the slope-intercept form of the equation of the line through the given points.

17) through: $(-4, 3)$ and $(3, -3)$

Write the standard form of the equation of each line given the slope and y-intercept.

18) Slope $= -\frac{3}{4}$, y-intercept $= 5$

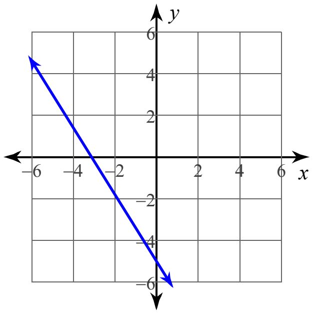
Write the standard form of the equation of each line.

19) $y = \frac{5}{2}x - 4$

Write the standard form of the equation of the line through the given points.

20) through: $(-2, -5)$ and $(-5, 4)$

Answers to HW # 7 Review #2 Part 1

1)


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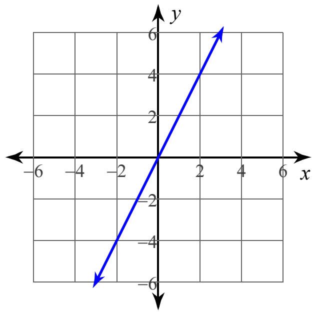

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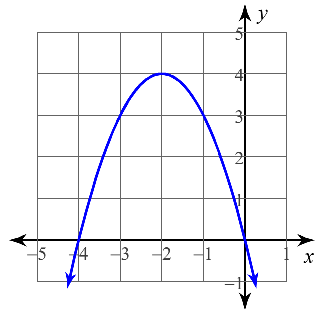

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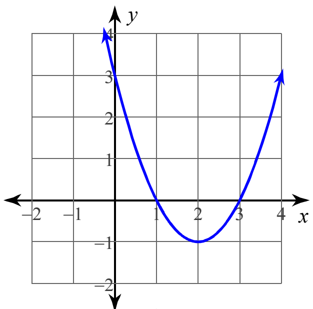

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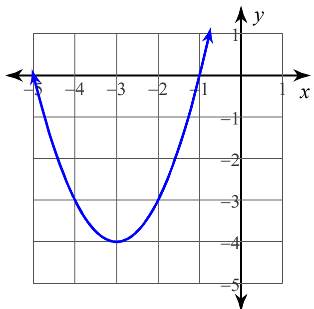

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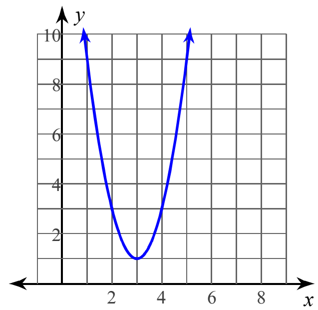

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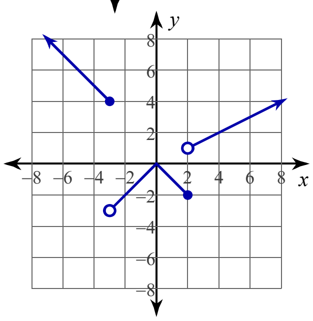

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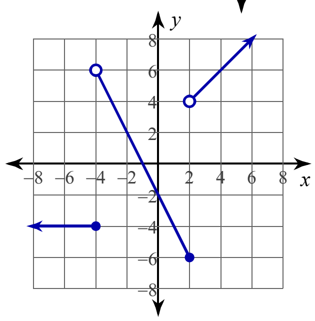

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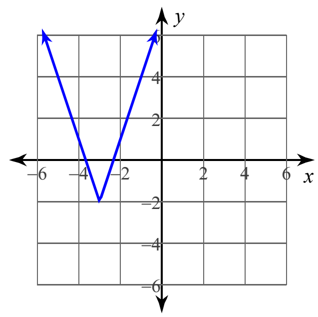

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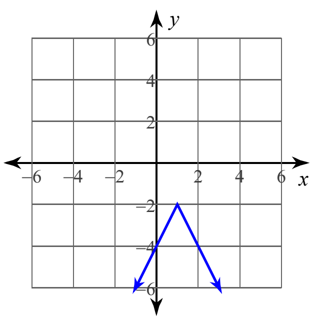

11) $\left\{1, -\frac{1}{5}\right\}$

12) $\{2, 3\}$

13) $1 < x < 8$

14) $-\frac{2}{3} \le p \le 2$

15) $y = -5x - 5$

16) $y = -\frac{5}{4}x + 2$

17) $y = -\frac{6}{7}x - \frac{3}{7}$

18) $3x + 4y = 20$

19) $5x - 2y = 8$

20) $3x + y = -11$